PHIL 120 Terms Sheet and Notes

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1 Arguments

Definition 1.1. (Conclusion Indicator Words) Words such as so, therefore, hence, thus, accordingly, consequently which indicate that the sentence that follows them is the conclusion of an argument.

Definition 1.2. (Premise Indicator Words) Words such as since, because, given that, for which indicate that the sentence(s) that follow are the premise(s) of an argument.

Definition 1.3. (Argument) A series of sentences that consists of some premises and a conclusion.

- When the premises are all presented before the conclusion, the argument is in **standard form**.

Definition 1.4. (Sentence) A propositional statement that can be either true or false.

- Note that questions (e.g. Are you sleepy yet?), imperatives/commands (e.g. Wake up!), and exclamations (e.g. Ouch!) are not sentences.

2 The Scope of Logic

Definition 2.1. (Valid Argument) An argument in which the conclusion follows from the premises (i.e. if the premises are true then the conclusion must also be true). Alternatively, an argument for which there are no counterexamples (see below).

Definition 2.2. (Invalid Argument) An argument in which the conclusion does not follow from the premises (i.e. a case can be found in which all the premises are true but the conclusion is false).

- A case in which all the premises are true but the conclusion is false is called a counterexample.

Definition 2.3. (Consequence) An sentence A is a consequence of sentence(s) B_1, B_2, $B_3 \ldots B_n$ if and only if there is no case where B_1, B_2, $B_3 \ldots B_n$ are all true and A is false. (i.e. there is no counterexample).

- We say that A follows from B_1, B_2, $B_3 \ldots$, B_n or that B_1, B_2, $B_3 \ldots$, B_n entail A.

Example 2.4. (Relationship between consequence and validity) Note that an argument is valid if and only if it is a consequence of the premises. An argument is invalid if and only if there is a counterexample (i.e. it is not valid).

Example 2.5. (Types of "Dependent" Validity) An argument is **Nomologically Valid** if there are no counterexamples that respect the laws of nature. An argument is **Conceptually Valid** if there no no counterexamples that respect the natural/conceptual meaning between words.

Definition 2.6. (Formally Valid Argument) An argument for which its validity only depends on its form/pattern, not on its content.

Example 2.7. An example of a formally valid argument is as follows:

- A is either an X or a Y
- A isn't a Y
- Therefore A is an X

Definition 2.8. (Sound Argument) An argument which is both valid and has premise(s) that are all true.

- Only sound arguments support their conclusion (i.e. if an argument is valid and supports its conclusion only if its premises are true).
- The converse it not true. Just because the premises and/or the conclusion are true does not necessarily mean that an argument is valid or supports its conclusion.

Definition 2.9. (Inductive Argument) An argument which is a generalization or an inference based on a set of premises.

- While an inductive argument may be supported by its premise(s), it is not valid because future counterexamples could be found.

Example 2.10. An example of an inductive argument is as follows:

- Every winter so far it has snowed in Calgary
- Therefore this upcoming winter it will snow in Calgary

Just because it has snowed in Calgary every winter so far does not mean it will snow in winter this upcoming winter. Thus this argument is inductive and invalid.

Question 2.11. (Could there be?)

1. An invalid argument that can be made valid by the addition of a new premise? Yes. We can add a contradiction to the premises which would prevent a counterexample from being found (since the premises could never all be T).
2. A valid argument that can be made invalid by the addition of a new premise? No. Assume this was the case. This would mean a counterexample must exist where all the premises (the new one and the old ones) are T. However, this would also mean that there was a case for which the original premises were all T but the conclusion F, hence the original argument was not valid. (Alternatively we can say that the cases allowed under the new set of premises are a "subset" of the cases allowed under the original ones, adding a new premise makes it harder for all premises to be true together)

Note that an argument being valid means that if follows from the underlying premise(s). Adding new premises does not change the fact that the conclusion has already been "proven" from the old ones. Hence, valid arguments always remain valid when new premises are added.

3 Other Logical Notions

Definition 3.1. (Jointly Possible) A set of sentences is jointly possible if and only if there is a case where they are all true together.

- If there is not case where all the sentences can be true together, the sentences are jointly impossible

Definition 3.2. (Necessary Truth) A sentence is a necessary truth if and only if it MUST be true (i.e. there is no case in which it is false).

Definition 3.3. (Necessary Falsehood) A sentence is a necessary falsehood if and only if it MUST be false (i.e. there is no case in which it is true).

Definition 3.4. (Contingent) A sentence is contingent if and only if it could be true or it could be false (i.e. it is not a necessary truth or false).

Example 3.5. The following are examples of necessary truth, necessary false, and contingent sentences

- (NT) It is either raining outside or it is not raining outside
- (NF) It is both raining outside and not raining outside
- (C) It is raining outside

Note that NTs and NFs can be deduced directly from the content of the sentence (i.e. they never require "exterior knowledge" and are self-evident). Contingent sentences are things that can be debated and/or disagreed upon (even if they are "facts" or happen to be true or false in reality). Something could have happened in an alternate universe that changed the truth value for all we know!

Definition 3.6. (Necessary Equivalence) Two sentences are necessarily equivalent if they always have the same truth value together. (i.e. if one is true the other is true, if one is false the other is false).

- Sentences with this property are said to be necessarily equivalent.

Question 3.7. (For each of the following are they a necessary truth, a necessary falsehood, or contingent?)

1. Caesar crossed the Rubicon. (C). For this could be false in an alternate universe.
2. Someone crossed the Rubicon. (C). Since possibly no one has ever crossed the Rubicon.
3. No one has ever crossed the Rubicon. (C). For someone could have crossed the Rubicon.
4. If Caesar has crossed the Rubicon then somebody has. (NT). This is a self-evident truth.
5. Even though Caesar crossed the Rubicon, nobody has. (NF). This is a self-evident falsehood.
6. If anyone has ever crossed the Rubicon, it was Caesar. (C). Someone else could has crossed as well.

Question 3.8. (For each of the following are they a necessary truth, a necessary falsehood, or contingent?)

1. Elephants dissolve in water. (C). For maybe they do not dissolve in water.
2. Wood is a light, durable substance for building things. (C). This could be false in an alternate universe.
3. If wood were a useful building material it would be good for building things. (NT). Self-evident truth.
4. I live in a three story building that is two stories tall. (NF). A self-evident falsehood.
5. If gerbils were mammals they would nurse their young. (C). Since in an alternate universe maybe mammals do not nurse their young.

Question 3.9. (Which of the following pairs of sentences are necessarily equivalent?)

1. Elephants dissolve in water / If you put an elephant in water it will disintegrate. (Equiv.). Since the sentences mean the same thing. They are either both T or both F.
2. All mammals dissolve in water / If you put an elephant in water it will disintegrate. (Not Equiv.). These sentences do not always have the same truth value. The first could be F while the second is T.
3. George Bush was the 43rd President / Barack Obama is the 44th President. (Not Equiv). Each sentence could be T or F independent of the other. They do not have to have the same truth value.
4. Barack Obama was the 44th President / Barack Obama was the President immediately after the 43rd President. (Equiv). These sentences mean the same thing. They are either both true or both false.
5. Elephants dissolve in water / All mammals dissolve in water. (Not Equiv). Follows directly from (1) and (2). These sentences do not need to have the same truth value. Elephants could dissolve in water while not all other mammals dissolve in water.

Question 3.10. (Which of the following pairs of sentences are necessarily equivalent?)

1. Theolonius Monk played piano / John Coltrane played tenor sax. (Not Equiv.). Since the sentences do not have to have the same truth value. Each can be T or F independent of the other.
2. Theolonius Monk played gigs with John Coltrane / John Coltrane played gigs with Theolonius Monk. (Equiv.). These sentences mean the same thing. They are either both T or both F.
3. All professional piano players have big hands / Piano player Bud Powell had big hands. (Not Equiv). The first sentence could be false while the second one is true.
4. Bud Powell suffered from severe mental illness / All piano players suffer from severe mental illness. (Not Equiv). The first sentence could be true while the second is false.
5. John Coltrane was deeply religous / John Coltrane viewed music as a deep expression of spirituality. (Not Equiv). Either sentence could be T or F independent of the other. Perhaps John Coltrane was religious but did not view music as an expression of spirituality.

Question 3.11. (Consider the following sentences:)

1. There are at least four giraffes at the wild animal park
2. There are exactly seven gorillas at the wild animal park
3. There are not more than two Martians at the wild animal park
4. Every giraffe at the wild animal park is a Martian

Consider the following collections of sentences. Which are jointly possible and which are jointly impossible?

1. (2, 3, 4) (JP). These sentences can all be T.
2. (1, 3, 4) (JI). Sentences 1 and 4 cannot be both T.
3. (1, 2, 4) (JP). These sentences can all be T.
4. (1, 2, 3) (JP). These sentences can all be T.

Question 3.12. (Consider the following sentences:)

1. All people are mortal
2. Socrates is a person
3. Socrates will never die
4. Socrates is mortal

Consider the following collections of sentences. Which are jointly possible and which are jointly impossible?

1. (1, 2, 3) (JI). Sentences 1 and 2 imply Socrates is mortal. This contradicts sentence 3 which says Socrates will never die.
2. (2, 3, 4) (JI). Sentences 3 and 4 contradict one another.
3. (2, 3) (JP). These sentences both be true T. Perhaps people don't die?
4. (1, 4) (JP). Evidently.
5. (1, 2, 3, 4) (JI). Sentences 3 and 4 contradict one another. Hence as a whole, these sentences are JI

Note that if any subset of a group of sentences is jointly impossible then the senteces taken as a whole are also jointly impossible.

Question 3.13. (Which of the following are possible? For each, if it is possible give an example. If it is not possible, explain why.)

1. A valid argument that has one false premise and one true premise. (Possible.). All ants are insects. All insects eat iron. Therefore ants eat iron.
2. A valid argument that has a false conclusion. (Possible.). I am 21 years old. I am also 22 years old. Thus birds can fly to Mars. (Note that since the premises are jointly impossible, the argument MUST be valid). Also see sol'n to above question.
3. A valid argument, the conclusion of which is a necessary falsehood. (Possible). As before, if the premises are jointly impossible, the argument is valid, even if the conclusion is a necessary falsehood.
4. An invalid argument, the conclusion of which is a necessary truth. (Not Possible). If the conclusion is a necessary truth, no counterexample can be found (which would require there to be a case in which the conclusion if false, all premises are true). Thus the argument MUST be valid.
5. A necessary truth that is contingent. (Not Possible). By definition this is not possible.
6. Two necessarily equivalent sentences, both of which are necessary truths. (Possible). Indeed ANY pair of necessary truths are necessarily equivalent since they always both have the same truth value (T).

7. Two necessarily equivalent sentences, one of which is a necessary truth and one of which is contingent. (Not possible). The necessary truth must always be true while the contingent can possibly be false (else it is a necessary truth). Thus they cannot be necessarily equivalent.

8. Two equivalent sentences that are jointly impossible. (Possible). For example if they are both necessary falsehoods (both always F) they are equivalent and jointly impossible. (E.g. It is both raining and not raining. It is neither raining nor not raining).

9. A jointly possible collection of sentences that contains a necessary falsehood. (Not possible). Since this directly implies that not all the sentences can be true at once.

10. A jointly impossible collection of sentences that contains a necessary truth. (Possible). If one of the other sentences is a necessary falsehood, the collection is automatically necessarily impossible.

Note that if the conclusion of an argument is necessarily true or if its premises are jointly impossible, it is always valid. Also note that if a collection of sentences has a necessary falsehood, it must be jointly impossible. NTs are always necessarily equivalent, NFs are always necessarily equivalent. Contingents can only be necessarily equivalent to other contingents.

Question 3.14. (Which of the following are possible? For each, if it is possible give an example. If it is not possible, explain why.)

1. A valid argument whose premises are all necessary truths and whose conclusion is contingent. (Not possible). Since the premises are NTs and the conclusion is C, this directly implies that a counterexample can be found (when the conclusion is false).

2. A valid argument with true premises and a false conclusion. (Not possible). For this would directly imply that there is a counterexample which would mean the argument is NOT valid.

3. A jointly possible collection of sentences that contains two sentences that are not necessarily equivalent. (Possible). Pick two contingent sentences. For example: All ants are insects. All ants are red.

4. A jointly possible collection of sentences, all of which are contingent. (Possible). See above.

5. A false necessary truth. (Not Possible). By definition this is not possible.

6. A valid argument with false premises. (Possible). All humans have horns. All horned beings have four legs. Therefore all humans have four legs.

7. A necessarily equivalent pair of sentences that are not jointly possible. (Possible). Just pick two necessary falsehoods.

8. A necessary truth that is also a necessary falsehood. (Not possible). By definition this is not possible.

9. A jointly possible collection of sentences that are all necessary falsehood. (Not possible). Since this directly implies all the sentences are never true and thus can't be true together.

4 First Steps to Symbolization

Several types of arguments can be valid simply based on their structure (and thus not on their content). Arguments with this property are said to be *valid by virtue of form*:

Example 4.1. Examples of arguments valid based on structure only:

- A
- If A then C
- Therefore C

- A or B
- not-A
- Therefore B

- not-(A and B)
- A
- Therefore not-B

Note that truth-functional logic (TFL) can only help us understand arguments that are formally valid. It cannot help us understand arguments that are conceptually valid.

Definition 4.2. (Truth-functional Logic or TFL) A formal language that can be used to symbolize formally valid arguments. NOTE: There are several nuances in the Engligh language that TFL cannot account for.

Definition 4.3. (Sentence Letter) Letters such as A, B, P_8, ..., that are used to represent atomic sentences in TFL. These are the building blocks for making more complex sentences.

- A symbolization key can be used to convert a sentence letter into an english meaning. Ex. A: It is raining outside, C: Jenny is miserable.

5 Connectives

Definition 5.1. (Connectives) Operations in TFL that represent logical and formal relationships between (logical) sentences. There are only five connectives:

Symbol	Name	Meaning
$\neg$	negation	'It is not the case that'
$\wedge$	conjunction	'Both ... and ...'
$\vee$	disjunction	'Either ... or ...'
$\rightarrow$	conditional	'If ... then ...'
$\leftrightarrow$	biconditional	'... if and only if ...'

5.1 Negation

Definition 5.2. (Negation) The negation operator can be used to represent the complement of a logical sentence. It represents the english word 'not'.

- A sentence can be symbolized as $\neg A$ if it can be paraphrase in English as 'it is not the case that'

Example 5.3. (Negation Operator) Examples of using the negation operator is as follows:

- R: The widget can be replaced $\implies$ $\neg R$: The widget cannot be replaced / The widget is irreplaceable
- C: I am in Calgary right now $\implies$ $\neg C$: I am not in Calgary right now

IMPORTANT!!! For a sentence such as J: Jane is happy, the negation $\neg J$ represents that state that Jane not happy, but **DOES NOT** represent Jane necessarily being unhappy

5.2 Conjunction

Definition 5.4. (Conjunction) The conjunction operator combines two logical sentences. It represents the english word 'and'.

- Two sentences that are the operands of the conjunction operator are said to be conjuncts
- A sentence can be symbolized as $(A \wedge B)$ if it can paraphrased as 'Both ... and ...' or as '... but ...' or as 'although..., ...'

Example 5.5. (Conjuction Operator) Assume that A: Barbara is athletic, B: Adam is athletic, and E: Barbara is energetic. Examples of using the conjunction operator are as follows:

- $(B \wedge E)$: Barbara is athletic and energetic
- $(A \wedge B)$: Adam and Barbara are both athletic
- $(E \wedge \neg B)$: Although Barbara is energetic, she is not athletic

IMPORTANT!!! The conjunction operator '$\wedge$' cannot be used to represent asymmetric differences (ex. temporal differences) since it is a symmetric (commutative) operator. For example, H_0: Harry stood up and objected to the proposal and H_1: Harry objected to the proposal and stood up, are two different sentences that are not the same under conjunction

5.3 Disjunction

Definition 5.6. (Disjunction) The conjunction operator combines two logical sentences. It represents the english word 'or'.

- Two sentences that are the operands of the disjunction operator are said to be disjuncts
- A sentence can be symbolized as $(A \vee B)$ if it can paraphrased as 'Either ... or ...'

Example 5.7. (Disjunction Operator) Assume that S_1: You will get soup, S_2: You will get salad. Examples of using the disjunction operator are as follows:

- $(\neg S_1 \vee \neg S_2)$: Either you will not get soup or not get salad
- $\neg(S_1 \vee S_2)$: You will neither get soup nor salad
- $(\neg S_1 \wedge S_2) \vee (S_1 \wedge \neg S_2)$: You will get either soup or salad but not both
- $(S_1 \vee S_2) \wedge \neg(S_1 \wedge S_2)$: You will get soup or salad but not both (same as above)

IMPORTANT!!! The disjunction operator '$\vee$' always represents the *inclusive or* operation. In other words it is true when either one or both of the disjuncts are true. It does not represent the *exclusive or* operator which is true when only one of the disjuncts is true (and false if neither or both are true).

5.4 Conditional

Definition 5.8. (Conditional) The conditional implicates two logical sentences. It represents 'If-then' relations from the English language in logical sentences.

- A sentence can be symbolized as $(A \rightarrow B)$ if it can paraphrased as 'If A then B' or 'A only if B'
- The first operand A is called the antecedent while the second operand B is called the consequent

Example 5.9. (Conditional) Assume that P: Jean is in Paris, F: Jean is in France. Examples of using the conditional are as follows:

- $(P \rightarrow F)$: For Jean to be in Paris it is necessary that Jean be in France
- $(P \rightarrow F)$: For John to be in France it is sufficient that he be in Paris (same as above)

IMPORTANT!!! The conditional operator tells us that if the antecedent is true then the consequent is true. It says nothing about the causal connection between the two events

5.5 Biconditional

Definition 5.10. (Biconditional) The biconditional marks the equivalence two logical sentences.

- A sentence can be symbolized as $(A \leftrightarrow B)$ if it can paraphrased as 'A iff B' or 'A if and only if B'
- Using the regular conditional operator, the biconditional can be written as $(A \rightarrow B) \wedge (B \rightarrow A)$

Example 5.11. (Conditional) Assume that D: Laika is a dog, M: Laika is a mammal:

- $(D \rightarrow M)$: Laika is a dog only if she is a mammal
- $(M \rightarrow D)$: Laika is a dog if she is a mammal
- $(D \leftrightarrow M)$: Laika is a dog if and only if she is a mammal

IMPORTANT!!! Be careful when using the biconditional. Make sure a normal condition isn't intended.

5.6 Unless

Definition 5.12. (Unless) A logical implication between two sentences.

- An 'unless' sentence can be symbolized as $(\neg A \to B)$ or $(\neg B \to A)$ or $(A \vee B)$

Example 5.13. (Unless) Assume that N: I will go running, R: It rains:

- $(\neg N \to R)$: If I don't go running then it rained
- $(\neg R \to N)$: If it does not rain then I will go running
- $(N \leftrightarrow \neg R)$: I will go running if and only if it does not rain

IMPORTANT!!! Unless is also a tricky word. Be careful to double check the meaning before you use it.

Question 5.14. (Using the symbolization key given, symbolize each English sentence in TFL?)

M : Those creatures are men in suits

C : Those creatures are chimpanzees

G : Those creatures are gorillas

1. Those creatures are not men in suits $\neg M$
2. Those creatures are men in suits or they are not. $(M \vee \neg M)$
3. Those creatures are either gorillas or chimpanzees. $(G \vee C)$
4. Those creatures are neither gorillas nor chimpanzees. $\neg(C \vee G)$
5. If those creatures are chimpanzees then they are neither gorillas nor men in suits. $C \rightarrow \neg(G \vee M)$
6. Unless those creatures are men in suits, they are either chimpanzees or gorillas. $\neg M \rightarrow (C \vee G)$ or $(M \vee (C \vee G))$

Question 5.15. (Using the symbolization key given, symbolize each English sentence in TFL?)

A : Mister Ace was murdered

B : Those butler did it

C : The cook did it

D : The duchess is lying

E : Mister Edge was murdered

F : The murder weapon was a frying pan

1. Either Mister Ace or Mister Edge was murdered $(A \vee E)$
2. If Mister Ace was murdered then the cook did it $(A \rightarrow C)$
3. If Mister Edge was murdered then the cook did not do it. $(E \rightarrow \neg C)$
4. Either the butler did it or the duchess is lying. $(B \vee D)$
5. The cook only did it only if the duchess if lying. $(C \rightarrow D)$
6. If the murder weapon was a frying pan then the culprit must have been the cook $(F \rightarrow C)$
7. If the murder weapon was not a frying pan than the culprit was either the cook or the butler $(\neg F \rightarrow (C \vee B))$
8. Mister Ace was murdered if and only if Mister Edge was not murdered $(A \leftrightarrow \neg E)$
9. The Duchess is lying, unless it was Mister Edge who was murdered $(D \vee E)$
10. If Mister Ace was murdered he was done in with a frying pan$(A \rightarrow F)$
11. Since the cook did it, the butler did not $(A \wedge \neg B)$
12. Of course the Duchess is lying! D

Question 5.16. (Using the symbolization key given, symbolize each English sentence in TFL?)

E_1 : Ava is an electrician

E_2 : Harrison is an electrician

F_1 : Ava is an firefighter

F_2 : Harrison is an firefighter

S_1 : Ava is satisfied with her career

S_2 : Harrison is satisfied with his career

1. Ava and Harrison are both electricians $(E_1 \wedge E_2)$
2. If Ava is a firefighter then she is satisfied with her career $(F_1 \rightarrow S_1)$
3. Ava is a firefighter unless she in an electrician. $(F_1 \vee E_1)$
4. Harrison is an unsatisfied electrician $(E_2 \wedge \neg S_2)$
5. Neither Ava nor Harrison is an electrician. $\neg(E_1 \vee E_2)$
6. Both Ava and Harrison are electricians but neither of them find it satisfying $((E_1 \wedge E_2) \wedge (\neg S_1 \wedge \neg S_2))$
7. Harrison is satisfied only if he is a firefighter $(S_2 \rightarrow E_2)$
8. If Ava is not an electrician, then neither is Harrison, but if she is then he is too $((\neg E_1 \rightarrow \neg E_2) \wedge (E_1 \rightarrow E_2))$
9. Ava is satisfied with her career if and only if Harrison is not satisfied with his $(S_1 \leftrightarrow \neg S_2)$
10. If Harrison is both an electrician and a firefighter, then he must be satisfied with his work $((E_2 \wedge F_2) \rightarrow S_2)$
11. It cannot be that Harrison is both an electrician and a firefighter $\neg(E_2 \wedge F_2)$
12. Harrison and Ava are both firefighters if and only if neither of them is an electrician $((F_2 \wedge F_1) \leftrightarrow \neg(E_2 \vee E_1))$

Question 5.17. (Using the symbolization key given, symbolize each English sentence in TFL?)

J_1 : John Coltrane played tenor sax

J_2 : John Coltrane played soprano sax

J_3 : John Coltrane played tuba

M_1 : Miles Davis played trumpet

M_2 : Miles Davis played tuba

1. John Coltrane played tenor sax and soprano sax $J_1 \wedge J_2$
2. Neither Miles Davis nor John Coltrane played tuba $\neg(J_3 \vee M_2)$
3. John Coltrane did not play both tenor sax and tuba $\neg(J_1 \wedge J_3)$
4. John Coltrane did not play tenor sax unless he also played soprano sax $(\neg J_1 \vee J_2)$
5. John Coltrane did not play tuba but Miles Davis did. $(\neg J_3 \wedge M_2)$
6. Miles Davis played trumpet only if he also played tuba $(M_1 \rightarrow M_2)$
7. If Miles Davis played trumpet, then John Coltrane played at least one of these three instruments: tenor sax, soprano sax, or tuba. $M_2 \rightarrow ((J_1 \vee J_2) \vee J_3)$
8. If John Coltrane played tuba then Miles Davis played neither trumpet nor tuba. $J_3 \rightarrow \neg(M_1 \vee M_2)$
9. Miles Davis and John Coltrane both played tuba if and only if Coltrane did not play tenor sax and Miles Davis did not play trumpet $(M_2 \wedge J_3) \leftrightarrow (\neg J_1 \wedge \neg M_1)$

Question 5.18. (Give a symbolization key and symbolize the following sentences.)

A : Alice is a spy

B : Bob is a spy

C : The code has been broken

G : The German embassy will be in an uproar

1. Alice and Bob are both spies $(A \wedge B)$
2. If either Alice or Bob is a spy, then the code has been broken. $((A \vee B) \rightarrow C)$
3. If neither Alice nor Bob is a spy, then the code remains unbroken. $(\neg(A \vee B) \rightarrow \neg C)$
4. The German embassy will be in an uproar, unless someone has broken the code $(G \vee C)$
5. Either the code has been broken or it has not, but the German embassy will be in an uproar regardless. $((C \vee \neg C) \wedge G)$
6. Either Alice or Bob is a spy, but not both. $((A \wedge \neg B) \vee (\neg A \wedge B))$

Question 5.19. (Give a symbolization key and symbolize the following sentences.)

F : There is food in the pridelands

R : Rafiki will talk about squashed bananas

A : Simba is alive

K : Scar will remain as king

1. If there is food to be found in the pridelands, then Rafiki will talk about squashed bananas. $(F \rightarrow R)$
2. Rafiki will talk about squashed bananas unless Simba is alive. $(R \vee A)$
3. Rafiki will either talk about squashed bananas or he won't, but there is food to be found in the pridelands regardless. $((R \vee \neg R) \wedge F)$
4. Scar will remain as king if and only if there is food to be found in the pridelands. $(K \leftrightarrow F)$
5. If Simba is alive, then Scar will not remain as king.. $(A \rightarrow \neg K)$

Question 5.20. (For each argument, write a symbolization key and symbolize all of the sentences of the argument in TFL. See pg. 47 for questions.)

1. D: Dorothy plays piano in the morning. R: Roger wakes up cranky. C: Dorothy is distracted.
   $D \rightarrow R$, $D \vee C$, $\therefore (\neg R \rightarrow C)$
2. R: It will rain on Tuesday. S: It will snow on Tuesday. D: Neville will be sad. C: Neville will be cold.
   $(R \vee S)$, $R \rightarrow D$, $S \rightarrow C$, $\therefore (D \vee C)$
3. Z: Zoog remembered to do his chores. C: Things are clean. N: Things are neat.
   $Z \rightarrow (C \wedge \neg N)$, $\neg Z \rightarrow (N \wedge \neg C)$, $\therefore (C \wedge \neg N) \vee (N \wedge \neg C)$

Question 5.21. (For each argument, write a symbolization key and symbolize all of the sentences of the argument in TFL. See pgs. 47/48 for questions.)

1. R: It will rain soon. L: My leg is hurting.
   L, $R \rightarrow L$, $\therefore R$
2. U: Doctor Octopus gets the uranium. B: Doctor Octopus blackmails the city. D: Doctor Octopus makes a dirty bomb.
   $U \rightarrow D$, $D \rightarrow B$, $\therefore (U \rightarrow B)$
3. B: Chinese government solves the weather shortages in Beijing. C: Chinese government moves the capital. Y: Chinese government diverts all water for the Yangtze river northward.
   $(\neg B \rightarrow C), \neg C, \therefore B$. $B \rightarrow Y, \therefore Y$.

Question 5.22. (How can we symbolize exclusive or using only two connectives? Is it possible to do it by only using one?.) XOR means that the two sentences are not equivalent (one must be true, but not both). Thus: $\neg(A \leftrightarrow B)$

6 Sentences of TFL

Definition 6.1. (Expression of TFL) Any string of logical symbols. These symbolic can include atomic sentences and sentence letters, connectives, and brackets. (Yes, this includes the gibberish too!)

Definition 6.2. (Sentence of TFL) The following statements define the sentences that are valid in TFL:

1. Every sentence letter is a sentence
2. If A is a sentence then $\neg A$ is a sentence
3. If A and B are sentences then each of $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, and $(A \leftrightarrow B)$ are sentences
4. Nothing else is a sentence

Note: This definition is an inductive definition because it starts with some base elements and provides a method to get indefinitely more elements by combining the base elements.

Definition 6.3. (Main Logical Operator) The last connective that was introduced into a logical sentence.

1. If the first operator in the sentence is '$\neg$' then that is the main logical operator
2. Otherwise start counting brackets, adding +1 for '(' and -1 for ')'. When the count is exactly 1, the first operator you hit (apart from '$\neg$') is the main logical operator

Note: If you use the above method for determining the main logical operator, you MUST include all the brackets (including the very outermost ones that enclose the entire sentence)

Definition 6.4. (Scope) The scope of a connective in a sentence is the subsentence for which that connective is the main logical operator

Question 6.5. (What are the rules for brackets in TFL?) In TFL, brackets play an important role in analyzing sentences. Thus, $(A \wedge B)$ is a sentence while $A \wedge B$ is not. That being said, there are some general rules and exceptions:

1. You can drop the outermost brackets of a sentence, so $(A \wedge B)$ is the same as $A \wedge B$. You must put the brackets back in if you embed or combine the sentence with another though.
2. You can use square brackets instead of parenthesis brackets, they have the same meaning. $[A \wedge B]$ is the same as $(A \wedge B)$

7 Ambiguity

Definition 7.1. (Ambiguous) A sentence is said to be ambiguous if it can have more than one meaning.

- In lexical ambiguity a sentence can contain words that have more than one meaning (e.g. bank of a river or bank as in financial institution)
- In structural ambiguity the same sentence can be interpreted in multiple ways (e.g. Flying planes can be dangerous)
- Scope ambiguities depend on whether one connective is in the scope of another (e.g. I like animals that are small and dangerous or stealthy)
- Ambiguity is different from vagueness. Something is vague when it is not possible to determine if it applies or not

In TFL, to avoid lexical ambiguity, use symbolization keys so as to disambiguate the words (e.g. by specifying (river) bank or (money) bank). Sentences in TFL cannot be structurally ambiguous. They must have only one meaning.

Example 7.2. (Scope Ambiguity) Consider the following examples of scope ambiguity:

– *Avengers: Endgame* is not long and boring. Let B: *Avengers: Endgame* is boring, L: *Avengers: Endgame* is long:

1. $\neg(L \wedge B)$: Avengers Endgame is not: both long and boring
2. $(\neg L \wedge B)$: Avengers Endgame is: not long, but is boring

– Tai Lung is either both small and dangerous or stealthy. Let D: Tai Lung is dangerous, S: Tai Lung is small, T: Tai Lung is stealthy:

1. $(S \wedge D) \vee T$: Tai Lung is both small and dangerous, or he is stealthy
2. $(S \wedge (D \vee T))$: Tai Lung is small, and he is also dangerous or stealthy

Question 7.3. (The following sentences are ambiguous. Give symbolization keys for each and symbolize the different meanings.)

1. Haskell is a birder and enjoys watching cranes H: Haskell is a birder. N: Haskell enjoys watching cranes (construction). N: Haskell enjoys watching cranes (bird).
   $H \wedge N$, $H \wedge B$

2. he zoo has lions or tigers and bears L: The zoo has lions. T: The zoo has tigers. B: The zoo has bears.
   $L \vee (T \wedge B)$, $(L \vee T) \wedge B$

3. The flower is not red or fragrant R: The flower is red. F: The flower is fragrant.
   $(\neg R \vee F)$, $\neg(R \vee F)$.

8 Use and Mention

Question 8.1. (Difference between a use and a mention)? When *using* a word, we are referring to the actual subject/object it is meant for (e.g. if it is someone's name, we are referring to the person and not the name). When we *mention* a word, we are referring to the word itself (even if it the name of something else).

Note: To remove the ambiguity between using and mentioning, we add quotation marks around mentions.

Example 8.2. (Use vs. Mention)

1. In the sentence "Aristotle was a founding father of logic" we are using the name Aristotle (no quotations since we are not mentioning his name).
2. In the sentence "'Aristotle' is a name that starts with a capital A and has 8 lower case letters," we are referring the name and not Aristotle himself (use quotations).

Definition 8.3. (Language Types) When we talk about a language, object language if the language we are talking about (it is the subject, e.g. TFL). The language we use to talk about the object language is the metalanguage (e.g. English).

Example 8.4. (Object Language and Subject Language) We must be careful when referring the object language within the subject language. Consider the sentences

1. D is a sentence letter of TFL
2. 'D' is a sentence letter of TFL

The first sentence technically invalid since we are referring (using) the TFL 'D' inside English (we are mixing to languages). In order to use 'D' in the English language we need to mention it, thus we wrap it in quotations as in the second sentence.

Definition 8.5. (Metavariables) Symbols in (augmented) English that can be used to mention arbitrary logical sentences. They are symbols in augmented English (and are not TFL symbols)

- While 'A' refers to the TFL (specific) sentence 'A' and that sentence only, the symbols $\mathcal{A}$, or $\mathcal{B}$, $\mathcal{C}$, etc... can be used to mention ANY (arbitrary) TFL sentence.

Definition 8.6. (Quotation Convention for Arguments). Suppose we would like to symbolize the premises and conclusion of an argument. We can do so by writing $\mathcal{A}_1, ..., \mathcal{A}_n \therefore \mathcal{E}$

Example 8.7. (Quotation Convention for Arguments). An example of an argument written using the Quotation Convention for Arguments is as follows: $A, A \to B \therefore B$. (A, A only if B, therefore B). Notice that no quotation marks were used, indicating that these symbols are being used and that this is a TFL sentence.

9 Characteristic Truth Tables

Example 9.1. (Truth Tables for Basic Connectives)

Negation

$\mathcal{A}$	$\neg\mathcal{A}$
T	F
F	T

Conjunction

$\mathcal{A}$	$\mathcal{B}$	$\mathcal{A} \wedge \mathcal{B}$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

$\mathcal{A}$	$\mathcal{B}$	$\mathcal{A} \vee \mathcal{B}$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional

$\mathcal{A}$	$\mathcal{B}$	$\mathcal{A} \rightarrow \mathcal{B}$
T	T	T
T	F	F
F	T	T
F	F	T

Bionditional

$\mathcal{A}$	$\mathcal{B}$	$\mathcal{A} \leftrightarrow \mathcal{B}$
T	T	T
T	F	F
F	T	F
F	F	T

Definition 9.2. (Material Conditional) The material conditional is the conditional operator that follows the above truth table (see Example 9.1). It is a non-commutative operator that is only true when the antecedent is true and the consequent is false.

You can think of the "conditional" as representing a statement in which someone is saying that one item follows from another. The only way such a statement can be false is if you find a counterexample (antecedent true, consequent false). Hence the reason why the conditional is only false in that singular case.

10 Truth-Functional Connectives

Definition 10.1. (Truth Functional) A connective is truth functional if and only if the truth value of a sentence with that connective as the main logical operator is determined only by the truth values of the constituent sentence(s).

Example 10.2. (Translating v. Symbolizing) We say that TFL symbolizes English sentences. This is because, while TFL can preserve truth values if sentences, it often cannot preserve subtleties (think sarcasm or emphasis or connotation) that sentences may contain.

11 Complete Truth Tables

Definition 11.1. (Valuation) A valuation is any assignment of truth values to particular sentences in TFL.

Definition 11.2. (Complete Truth Table) A truth table that has line for every possible valuation of the relevant letters in a sentence is said to be complete. In general, a complete truth table for a sentence with N sentence letters will have 2^N rows.

IMPORTANT ASIDE: ALWAYS, ALWAYS, ALWAYS WRITE ALL THE BRACKETS. ALWAYS. DOESN'T MATTER IF THE OPERATORS ARE COMMUTATIVE. WRITE THE BRACKETS!

12 Semantic Concepts

Definition 12.1. (Tautology) A tautology is a sentence that is true on every valuation. This is the idea/extension/surrogate of a "necessary truth" to TFL.

Definition 12.2. (Contradiction) A contradiction is a sentence that is false on every valuation. This is the idea/extension/surrogate of a "necessary falsehood" to TFL.

Definition 12.3. (Equivalence) $\mathcal{A}$ and $\mathcal{B}$ are said to be equivalent if and only if their truth values are the same on every valuation (i.e. there is no valuation in which they have different truth values).

Definition 12.4. (Jointly Satisfiable) $\mathcal{A}_1, \mathcal{A}_2, ...\mathcal{A}_n$ are said to be jointly satisfiable if there is a valuation for which they are all true. This is the TFL surrogate for joint possiblity.

- On a similar token, if there is not valuation for which all the sentences are true then they are said to be jointly unsatisfiable

Definition 12.5. (Entailment) Sentences $\mathcal{A}_1, \mathcal{A}_2, ...\mathcal{A}_n$ entail (in TFL) the sentence $\mathcal{C}$ if and only if no valuation of the relevant sentences makes $\mathcal{A}_1, \mathcal{A}_2, ...\mathcal{A}_n$ all true and $\mathcal{C}$ false.

Example 12.6. (Double Turnstile and Entailment)

To represent entailment in the "metalanguage" we can use the double turnstile $\vDash$. Thus, the following argument $\mathcal{A}, \mathcal{A} \to \mathcal{B} \vDash \mathcal{B}$ can be translated as "The TFL sentences $\mathcal{A}$ and $\mathcal{A} \to \mathcal{B}$ entail $\mathcal{B}$".

Note that to show that sentence $\mathcal{C}$ is a tautology we write $\vDash \mathcal{C}$. To show something is a contradiction we can write $\mathcal{C} \vDash$. To show (for example) that sentences $\mathcal{A}_1, \mathcal{A}_2, ...\mathcal{A}_n$ do not entail $\mathcal{C}$ we can write $\mathcal{A}_1, \mathcal{A}_2, ...\mathcal{A}_n \nvDash \mathcal{C}$

13 Limitations of TFL

Example 13.1. (Limitations of TFL) TFL is not capable analyzing english arguments that are true/false based on experince. For example, the argument "Daisy has four legs, therefore Daisy has more than two legs", while true, would require two sentences to be symbolized in TFL.

Mixed-truths such as the sentence, "Jan is neither bald nor not bald" cannot be symbolized in TFL. This is because this sentence reduces to $\neg J \wedge \neg\neg J$ which is a contradiction. It could be true that Jan is partially bald, however.

Finally, consider the argument, "If God exists, He does not answer malevolent prayers". You may think to symbolize this is $\neg(G \rightarrow M)$, however this would (somehow, check this) entail G. This is not correct, however, since even an atheist could agree with that sentence (and yet still not believe in God). A more 'correct' argument (at least for TFL) would be to say that "If God exists, She does not answer malevolent prayers" leading to a symbolization of $G \rightarrow \neg M$. While this fixes the entailment issue, note that now if G is false then both the sentences $G \rightarrow \neg M$ and $G \rightarrow M$ are true, leading to another contradiction! (see *paradox of the material conditional*)

14 Truth Table Shortcuts

Example 14.1. (Truth Table Calculation Speedups)

1. When calculating disjuntions, you automatically know it is T if either of the disjuncts is true
2. When calculating conjunctions you automatically know it is F if either of the conjuncts is false.
3. When calculating conditionals you automatically know the conditional is T if either the consequent is T or if the antecedent if F.

15 Partial Truth Tables

Example 15.1. (Partial Truth Tables) In order to check whether sentence(s) fit certain definitions, we can sometimes use partial truth tables:

Term	**Yes (Prove)**	**No (Disprove)**
Tautology?	Complete	One-line partial
Contradiction?	Complete	One-line partial
Equivalent?	Complete	One-line partial
Satisfiable?	One-line partial	Complete
Valid?	Complete	One-line partial
Entailment?	Complete	One-line partial

16 The Very Idea of Natural Deduction

Example 16.1. (Truth Tables vs. Proofs)

Using truth tables can very quickly become *intractable*. For example, a truth table for sentences with 10 sentence letters would require 1024 rows. In addition, truth tables don't give a lot of insight into why a sentence may be true or false. Hence, we create a proof system.

17 Basic Rules for TFL

REMEMBER: ALWAYS MAKE SURE TO WRITE PR ON THE PROOF PREMISES (IF ANY)

17.1 Reiteration

Definition 17.1. (Reiteration, pg. 115) The reiteration rule lets you write a line again that you have already previously written before.

MAKE SURE TO WRITE [R, m] ON LINE n WHERE m IS WHERE THE INITIAL LINE OCCURRED

17.2 Conjunction

Definition 17.2. (Conjunction Introduction, pg. 117) If you have lines $\mathcal{A}$ and $\mathcal{B}$ then you can get line $\mathcal{A} \wedge \mathcal{B}$ on any subsequent lines.

MAKE SURE TO WRITE [$\wedge I$, m, n] WHERE m, n ARE THE LINES OF THE CONJUNCTS

Definition 17.3. (Conjunction Elimination, pg. 117) If you have line $\mathcal{A} \wedge \mathcal{B}$ then you can get either line $\mathcal{A}$ or $\mathcal{B}$ on any subsequent lines.

MAKE SURE TO WRITE [$\wedge E$, m] WHERE m IS THE LINE OF THE CONJUNCTION

17.3 Conditional

Definition 17.4. (Conditional Elimination, pg. 121) If you have line $\mathcal{A} \rightarrow \mathcal{B}$ and $\mathcal{A}$ then you can get $\mathcal{B}$ on any subsequent lines.

MAKE SURE TO WRITE [$\rightarrow E$, m, n] WHERE m IS THE LINE OF THE CONDITIONAL AND N IS THE LINE OF THE ANTECEDENT

Definition 17.5. (Conditional Introduction, pg. 123) If you can assume $\mathcal{A}$ in a subproof and use only previous lines (within the subproof or other proofs/subproofs that ENCLOSE the current one) to derive $\mathcal{B}$ then you can get $\mathcal{A} \rightarrow \mathcal{B}$ in the directly superior proof/subproof.

MAKE SURE TO WRITE [$\rightarrow I$, m-n] WHERE m-n ARE THE LINES OF THE SUBPROOF OF THE CONDITIONAL. MAKE SURE TO WRITE [AS] IN THE SUBPROOF ASSUMPTIONS

17.4 Biconditional

Definition 17.6. (Bionditional Introduction, pg. 131) If you can assume $\mathcal{A}$ in a subproof to derive $\mathcal{B}$ and use $\mathcal{B}$ in a subproof to derive $\mathcal{A}$ then you can get $\mathcal{A} \leftrightarrow \mathcal{B}$ in the directly superior proof/subproof.

MAKE SURE TO WRITE [$\leftrightarrow I$, i-j, k-l] WHERE m-n and k-l ARE THE LINES OF THE RESPECTIVE SUBPROOFS. MAKE SURE TO WRITE [AS] IN EACH SUBPROOF ASSUMPTION

Definition 17.7. (Biconditional Elimination, pg. 131) If you have the lines $\mathcal{A} \leftrightarrow \mathcal{B}$ and $\mathcal{A}$ then you can get $\mathcal{B}$ on any subsequent lines.

MAKE SURE TO WRITE [$\leftrightarrow E$, m,n] WHERE m and n are the lines of the biconditional and the sentence. Write the biconditional first.

17.5 Disjunction

Definition 17.8. (Disjunction Introduction, pg. 132) If you have the line $\mathcal{A}$ then you can get $\mathcal{A} \vee \mathcal{B}$ on any subsequent lines where $\mathcal{B}$ is any sentence.

MAKE SURE TO WRITE [$\vee I$, m] WHERE m IS THE LINE OF THE ORIGINAL DISJUNCT

Definition 17.9. (Disjunction Elimination, pg. 133) If you have the line $\mathcal{A} \vee \mathcal{B}$ and you can use $\mathcal{A}$ to derive $\mathcal{C}$ in a subproof as well as $\mathcal{B}$ to derive $\mathcal{C}$ in a subproof then you can assume $\mathcal{C}$ on any subsequent lines.

MAKE SURE TO WRITE [$\vee E$, m, i-j, k-l] WHERE m IS THE LINE OF THE DISJUNCTION and i-j, k-l ARE THE LINES OF THE SUBPROOFS. MAKE SURE TO WRITE [AS] FOR EACH SUBPROOF ASSUMPTION.

17.6 Negation

Definition 17.10. (Negation Elimination, pg. 135) If you have the line $\mathcal{A}$ and the line $\neg\mathcal{A}$ you can introduce the line $\bot$.

MAKE SURE TO WRITE [$\neg E$, m, n] WHERE m and n ARE THE LINES OF THE TWO NEGATIVES. CONVENTION IS THE WRITE THE LINE WITH THE NEGATION FIRST, THEN THE LINE IT IS THE NEGATION OF.

Definition 17.11. (Negation Introduction, pg. 136) If you can assume the line $\mathcal{A}$ in a subproof and derive the $\bot$ then you can introduce the line $\neg\mathcal{A}$ in the direct outer proof/subproof.

MAKE SURE TO WRITE [$\neg I$, i-j] WHERE i-j ARE THE LINES OF THE SUBPROOF WHERE THE CONTRADICTION WAS DERIVED. MAKE SURE TO WRITE [AS] IN THE SUBPROOF ASSUMPTIONS.

Definition 17.12. (Indirect Proof, pg. 137) If you can assume the line $\neg\mathcal{A}$ in a subproof and derive the $\bot$ then you can introduce the line $\mathcal{A}$ in the direct outer proof/subproof.

MAKE SURE TO WRITE [IP, i-j] WHERE i-j ARE THE LINES OF THE SUBPROOF WHERE THE CONTRADICTION WAS DERIVED. MAKE SURE TO WRITE [AS] IN THE SUBPROOF ASSUMPTIONS.

Definition 17.13. (Explosion, pg. 139) If you can derive $\bot$ then any line $\mathcal{A}$ can be written on the subsequent line (you can say anything).

MAKE SURE TO WRITE [X, m] WHERE m IS THE LINE OF THE CONTRADICTION.

18 Constructing Proofs

Example 18.1. (Constructing Proofs) When constructing proofs, two strategies are to work forwards from a conclusion/given statement or to work backwards from a given statement. See Chapter 18 for more details.

REMEMBER: ALWAYS MAKE SURE TO WRITE PR ON THE PROOF PREMISES (IF ANY). ALSO WRITE [AS] ON SUBPROOF ASSUMPTIONS. ALWAYS CITE RULES THAT ARE BEING USED.

19 Additional Rules for TFL

This section is N/A.

20 Constructing Proofs

Definition 20.1. (Single Turnstile) If we have the statement $\mathcal{A}_\infty, \mathcal{A}_\in, ...\mathcal{A}_\setminus \vdash \mathcal{C}$ then it means we have a proof whose undisclosed assumptions are among $\mathcal{A}_\infty, \mathcal{A}_\in, ...\mathcal{A}_\setminus$.

- If we want to say that there is not a subproof which ends with $\mathcal{C}$ having $\mathcal{A}_\infty, \mathcal{A}_\in, ...\mathcal{A}_\setminus$ as assumptions then we can say $\mathcal{A}_\infty, \mathcal{A}_\in, ...\mathcal{A}_\setminus \nvdash \mathcal{C}$

Definition 20.2. (Theorem) We can say the sentence $\mathcal{A}$ is a theorem if and only if $\vdash \mathcal{A}$ (i.e. we can prove $\mathcal{A}$ with no undischarged assumptions).

Definition 20.3. (Interderivable) We can say two sentences are interderivable if they can both be proven from one another, i.e., $\mathcal{A} \vdash \mathcal{B}$ and $\mathcal{B} \vdash \mathcal{A}$.

Definition 20.4. (Jointly Inconsistent) We can say a set of sentences $\mathcal{A}_\infty, \mathcal{A}_\in, ...\mathcal{A}_\setminus$ are jointly inconsistent if and only if you can prove the contradiction $\bot$ from them.

- If a group of sentences is not inconsistent then they are said to be jointly consistent.

Example 20.5. (Concept Proof/Disproof Table)

Term	**Yes (Prove)**	**No (Disprove)**
Theorem?	One proof	All possible proofs
Inconsistent?	One proof	All possible proofs
Equivalent?	Two proofs	All possible proofs
Consistent?	All possible proofs	One proof

21 Derived Rules

This section is N/A.

22 Soundness and Completeness

Definition 22.1. (Soundness) The argument from $\vdash$ to $\models$ is the question of soundness. An proof system is said to be sound if there are no derivations that can be proved invalid by truth tables. It means that $\mathcal{A} \vdash \mathcal{B}$ implies $\mathcal{A} \models \mathcal{B}$.

- This involves showing that every TFL rule preserves the property that its conclusion is entailed by its premises as well as open assumptions

Definition 22.2. (Completeness) The argument from $\models$ to $\vdash$ is the question of completeness. An proof system is said to be complete if and only if there is a derivation of every semantically sound argument. It means that $\mathcal{A} \models \mathcal{B}$ implies $\mathcal{A} \vdash \mathcal{B}$.

- This means that your proof system includes all the rules that it needs to and have not left any out.

NOTE: THE PROOF SYSTEM IN TFL IS BOTH SOUND AND COMPLETE

Concepts	**Truth table (semantic) definition**	**Proof-theoretic (syntactic) definition**
Tautology/theorem	A sentence whose truth table only has Ts under the main connective	A sentence that can be derived without any premises.
Contradiction/inconsistent sent.	A sentence whose truth table only has Fs under the main connective	A sentence from which the contradiction $\bot$ can be derived
Contingent sent.	A sentence whose truth table contains both Ts and Fs under the main connective	A sentence that is not a theorem or inconsistent
Equivalent/ interderivable sent.	The columns under the main connectives are identical.	The sentences can be derived from each other
Unsatisfiable/inconsistent sent.	Sentences which do not have a single line in their truth table where they are all true.	Sentences from which one can derive the contradiction $\bot$
Satisfiable/Consistent sent.	Sentences which have at least one line in their truth table where they are all true.	Sentences from which the contradiction $\bot$ cannot be derived
Valid argument	An argument whose truth table has no lines where there are all Ts under main connectives for the premises and an F under the main connective for the conclusion.	An argument where one can derive the conclusion from the premises.

23 Building Blocks of FOL

Definition 23.1. (Name) In FOL, names are lowercase letters 'a' through 'r'. In order to use the same letter more than once, you can add subscripts.

- In FOL, each name must pick out exactly one item. There cannot be ambiguity. Different names can pick out the same item though.

Definition 23.2. (Predicates) In FOL predicates are capital letters 'A' through 'Z' with/without subscripts.

Example 23.3. (Quantifiers) Examples of quantifiers include the following statements:

1. A(x) $_____x$ is angry
2. H(x) $_____x$ is happy
3. B(x) Vinnie borrowed $_____x$ from Nunzio

Definition 23.4. (Universal Quantifier) The universal quantifier $\forall$ is a quantifier that says a predicate true for all members of a domain. (Think of it as an "all" statement on an array in Python where the predicate has been applied to all objects)

Example 23.5. (Universal Quantifier) Assume that the predicate H(x): $_____x$ is happy. Then the universal quantifier $\forall x H(x)$ means that everyone in the domain is happy.

Definition 23.6. (Existential Quantifier) The existential quantifier $\exists$ is a quantifier that says a predicate is true for at least one member of a domain. (Think of it as an "any" statement on an array in Python where the predicate has been applied to all objects)

Example 23.7. (Universal Quantifier) Assume that the predicate H(x): $_____x$ is happy. Then the universal quantifier $\exists x H(x)$ means that at least one person in the domain is happy.

NOTE: Whenever using a quantifier, there must be a variable associated with it.

Example 23.8. (Quantifers) Assume that $H(x)$: $_____x$ is happy. Then:

1. $\forall x H(x)$ means everybody is happy
2. $\exists x H(x)$ means at least one person is happy
3. $\neg\forall x H(x)$ and means not everyone is happy
4. $\exists x \neg H(x)$ means someone is unhappy (same as above)

NOTE: In general, $\neg\forall x \mathcal{H}(x)$ and $\exists x \neg \mathcal{H}(x)$ are equivalent (for all predicates)

Definition 23.9. (Domains) In FOL, a domain is a collection of items we are talking about. A domain must always have at least one member.

- The range of quantifiers is over the domain of items we have defined
- A member of the domain may be picked out by one name, many names, or no names, however every name must pick out one member of the domain.

24 Sentences with one Quantifier

Definition 24.1. (Every) In FOL, a sentence can be paraphrased as every F is a G if it is in the form $\forall x(\mathcal{F}(x) \to \mathcal{G}(x))$

Definition 24.2. (Some) In FOL, a sentence can be paraphrased as some F is a G if it is in the form $\exists x(\mathcal{F}(x) \wedge \mathcal{G}(x))$

Definition 24.3. (No) In FOL, a sentence can be paraphrased as no F is a G if it is in the form $\neg\exists x(\mathcal{F}(x) \wedge \mathcal{G}(x))$ or alternatively $\forall x(\mathcal{F}(x) \to \neg\mathcal{G}(x))$

Definition 24.4. (Only) In FOL, a sentence can be paraphrased as only F's are G's if it is in the form $\neg\exists x(\mathcal{G}(x) \wedge \neg\mathcal{F}(x))$ or alternatively $\forall x(\mathcal{G}(x) \to \mathcal{F}(x))$

Example 24.5. (Quantifier Paraphrases) Consider the domain all coins and the the predicates:

- $P(x)$: $_____x$ is in my pocket
- $T(x)$: $_____x$ is on the table
- $Q(x)$: $_____x$ is a quarter
- $D(x)$: $_____x$ is a dime

The the following mean:

1. $\forall x(P(x) \to Q(x))$: Every coin in my pocket is a quarter
2. $\exists x(T(x) \wedge D(x))$: Some coin on the table is a dime
3. $\neg\forall x(T(x) \to D(x))$ or $\exists x(T(x) \wedge \neg D(x))$: Not all coins on the table are dimes
4. $\neg\exists x(P(x) \wedge D(x))$ or $\forall x(P(x) \to \neg D(x))$: No coins in my pocket are dimes
5. $\neg\exists x(T(x) \wedge \neg D(x))$ or $\forall x(T(x) \to D(x))$: Only dimes are on the table

Definition 24.6. (Empty Predicates) A predicate that applies to nothing in the domain is called an empty predicate.

- When $\mathcal{F}$ is an empty predicate then the statement $\forall x(\mathcal{F} \to ...)$ is (vacuously) true.

Example 24.7. (Empty Predicates) Consider the predicates:

- $M(x)$: $_____x$ is a monkey
- $S(x)$: $_____x$ knows sign languge
- $R(x)$: $_____x$ is a refrigerator

Then the following can be symbolized as:

1. Every monkey knows sign language : $\forall x(M(x) \to S(x))$
2. Some monkey knows sign language : $\exists(M(x) \wedge S(x))$
3. Every refrigerator is a monkey : $\forall x(R(x) \to M(x))$

Note that the statement (1) can be true and statement (2) can be false at the same time, assuming the domain has no monkeys. Hence statement (2) can be *vacuously true*.

Similarly, assuming our domain consists of animals, statement (3) is *vacuously true* because there are no refrigerators in the domain.

Example 24.8. (Scope) The scope of quantifers follows the same rules as the scope for negations. Consider the following example.

- If everyone is a bassist then Lars is a bassist : $\forall x B(x) \to B(l)$ (this is necessarily true)
- Everyone is such that if they are a bassist then Lars is a bassist : $\forall x(B(x) \to B(l))$ (this is not necessarily true)
- If anyone is a bassist then Lars is a bassist : $\exists x B(x) \to B(l)$

NOTE: the statements $\forall x(\mathcal{F}(x) \to \mathcal{F}(l))$ and $\exists x \mathcal{F}(x) \to \mathcal{F}(l)$ are equivalent in general

Example 24.9. (Ambiguity) Be careful with ambiguity. For example, if you have a sentence like "Carol is a skilled surgeon and a tennis player" your predicate cannot just be skilled because it does not distinguish between being a skilled tennis player vs. being a skilled surgeon. You must use "skilled surgeon" as the predicate.

NOTE: In general, be very careful with predicates and adjectives.

25 Multiple Generality

Definition 25.1. (One-Place Predicate) A one place predicate is a predicate that has only one gap

Definition 25.2. (Two-Place Predicate) A two place predicate is a predicate that has two gaps

Definition 25.3. (Three-Place Predicate) A three place predicate is a predicate that has three gaps

Example 25.4. (N-place Predicates) Consider the following examples:

- $H(x)$: $_____x$ is happy (one-place)
- $L(x,y)$: $_____x$ loves $_____y$ (two-place)
- $B(x,y,z)$: $_____x$ borrowed $_____y$ from $_____z$ (three-place)

NOTE: When using multiple predicates, pay attention to the gaps and the subscripts on the gaps!

Example 25.5. (Quantifer Order) Pay attention to quantifer orders. They are not commutative. For example, consider the sentence "everyone loves someone". This can mean two things (Assume $L(x,y)$: $_____x$ loves $_____y$):

- For every person x, there is a person who x loves : $\forall x \exists y L(x,y)$
- There is a particular person that everybody loves : $\exists y \forall x L(x,y)$

These do not mean the same thing. Shifting them around is the *quantifier shift fallacy*.

Example 25.6. (Symbolizing Sentences) Consider the following predicates and symbolization key:

- domain : people and dogs
- $D(x)$: $_____x$ is a dog
- $F(x,y)$: $_____x$ is a friend of $_____y$
- $O(x,y)$: $_____x$ owns $_____y$
- g : Geraldo

Then the following senteces can be symbolized as:

1. Geraldo is a dog owner : $\exists x D(x) \wedge O(g,x)$
2. Someone is a dog owner : $\exists y \exists x D(x) \wedge O(y,x)$
3. All of Geraldo's friends are dog owners ; $\forall x(F(x,g) \rightarrow \exists z(D(z) \wedge O(x,z)))$
4. Every dog owner is a friend of a dog owner : $\forall x[\exists z(D(z) \wedge O(x,z)) \rightarrow \exists y(\exists z(D(z) \wedge O(y,z)) \wedge F(x,y))]$
5. Every dog owner's friend owns a dog of a friend :
   $\forall x[\exists y(F(x,y) \wedge \exists z(D(z) \wedge O(y,z))) \rightarrow \exists y((D(y) \wedge O(x,y)) \wedge \exists z(F(z,x) \wedge O(z,y)))]$

‘

Example 25.7. (Everyone, someone, no one) When using the words everyone, someone, and no one you may have to add a predicate $P(x)$: $_____x$ is a person. To make it correct. For example:

- Everyone is a friend of Geraldo : $\forall x(P(x) \rightarrow F(x,g))$
- No one is a friend of Geraldo : $\forall x(P(x) \rightarrow \neg F(x,g))$
- Someone is a friend of Geraldo : $\exists x(P(x) \wedge F(x,g))$

NOTE: That your symbolizations should represent what the sentence says though. So in cases that do not use the words someone, no one, everyone, you do not need to assume personhood.

26 Identity

This section is N/A.

27 Sentences of FOL

Definition 27.1. (Expression of FOL) an expression of FOL is any string of symbols of FOL. There are six types of symbols of FOL:

- Predicates: A, B, C, ... Z with/without subscripts
- Names: a, b, c, ..., r with/without subscripts
- Variable: s, t, u, ... z, with/without subscripts
- Connectives: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$
- Brackets: (,)
- Quantifers: $\forall, \exists$

Definition 27.2. (Term) A term is any name or variable.

- Examples of terms include a, b, x, x_1, x_2, y, y_{254}, z, etc.

Definition 27.3. (Atomic Formulas) An atomic formula satisfies the following:

1. Any sentence letter is an atomic formula
2. If $\mathcal{R}$ is an n-place predicate an $t_1, t_2, ...t_n$ are terms then $\mathcal{R}(t_1, t_2, ..., t_n)$ is an atomic formula
3. If t_1 and t_2 are terms then $t_1 = t_2$ is an atomic formula
4. Nothing else is an atomic formula

NOTE: Since sentence letters are also atomic formulas, all TFL sentences are formulas of FOL. The contradiction $\perp$ is also an atomic formula of FOL.

Definition 27.4. (Formula of FOL) Formulas of FOL are the following:

1. Every atomic formula is a formula
2. If $\mathcal{A}$ is a formula then $\neg\mathcal{A}$
3. If $\mathcal{A}$ and $\mathcal{B}$ are formulas then $(\mathcal{A} \circ \mathcal{B})$ is formula where $\circ$ is any of the TFL connectives $\wedge, \vee, \rightarrow, \leftrightarrow$
4. If $\mathcal{A}$ is a formula and x is a variable then $\forall x\mathcal{A}$ is a formula
5. If $\mathcal{A}$ is a formula and x is a variable then $\exists x\mathcal{A}$ is a formula
6. Nothing else is a formula

Definition 27.5. (Main Logical Operator) The main logical operator in a formula is the operator that was introduced most recently when the formula was introduced using the recursion rules.

Definition 27.6. (Scope) The scope of a logical operator is the sub-formula for which that operator is the main logical operator.

Definition 27.7. (Bound) A variable is said to be bound if it falls within the scope of either $\forall x$ or $\exists x$.

- An occurrence of a variable that is not bound is said to be free.

Example 27.8. Consider the formula $(\forall x(E(x) \vee D(y)) \rightarrow \exists z(E(x) \rightarrow L(z, x)))$. The first x and the second z are bound variables. The y and the second and third x's are free variables.

Definition 27.9. (Sentence) A sentence of FOL is any formula of FOL that contains no free variables.

NOTE: From the above definitions that you can still apply the quantifiers on a bound variable to obtain a sentence of FOL. For example $\exists x(\forall x F(x) \to G(x))$ is a valid sentence of FOL. It could be rewritten as $\exists x(\forall y F(y) \to G(x))$ for cleanliness though.

Also note that a bound variable is also bound by the quantifer that is closest to it with a matching variable. By closest, we mean the quantifer with the smallest sub-formula for which it is the main logical operator that encloses/captures/encapsulated the specific occurrence of the variable in question.

Example 27.10. (Vacuous Quantification) Note that the rules do not require our variable to occur free in $\mathcal{A}$ in order for $\forall x\mathcal{A}$ or $\exists x\mathcal{A}$ to be a formula of FOL.

For example $\forall x F(a)$ and $\exists x \forall x G(x)$ are both valid formulas of FOL. This is called *vacuous quantification*. These are respectively equivalent to $F(a)$ and $\forall x G(x)$ since the outer x does not bind in either case.

Example 27.11. (Long Conjunctions and Disjuctions). For long conjunctions and disjunctions you may leave out the inner brackets (as long as they are only conjuctions/disjunctions). Thus $(A \wedge B \wedge C \wedge D...)$ is a valid conjunction. And also $(A \vee B \vee C \vee D...)$ is a valid disjuction.

You MUST enclose the conjuction/disjuction with brackets though (unless they are the outermost brackets). You also cannot mix conjuctions/disjuctions with each other or other connectives. Thus $A \vee B \wedge C \wedge D$ and $B \vee C \to D$ are invalid formulas.

28 Definite Descriptions

This section is N/A.

29 Ambiguity

There is no way to sort out scope ambiguity in FOL because all quantifiers and connectives have well-defined scopes.

Example 29.1. Consider the saying "Everything that glitters is not gold" idiom. In FOL, this is an ambiguous sentence:

- Let $F(x)$: $_____x$ glitters and $G(x)$: $_____x$ is gold.
- Nothing that glitters is gold : $\forall x(F(x) \to \neg G(x))$ or $\neg\exists x(F(x) \wedge G(x))$
- Not everything that glitters is gold : $\neg\forall x(F(x) \to G(x))$ or $\exists x(F(x) \wedge \neg G(x))$

Example 29.2. Consider the sentence "Only young cats are playful":

- If something is playful then it is young and a cat : $\forall x(P(x) \to (Y(x) \wedge C(x)))$
- The only cats that are playful are the young ones : $\forall x((C(x) \wedge P(x)) \to Y(x))$
- The only animals that are young and playful are cats : $\forall x((Y(x) \wedge P(x)) \to C(x))$

Example 29.3. Consider the sentence "Everyone went to see a movie":

- Let $P(y)$: $_____y$ is a person. $M(x)$: $_____x$ is a movie. $S(y,x)$: $_____y$ went to see $_____x$. domain: people and movies
- There exists a single movie that every person went to see : $\exists x(M(x) \wedge \forall y(P(y) \to S(y,x)))$
- Every person went to see some move (not necessarily identical) : $\forall y(P(y) \to \exists x(M(x) \wedge S(y,x)))$

In the first reading we say that the existential quantifier has *wide scope* since its scope encloses the universal quantifier. In the second reading the universal quantifier has wide scope since its scope encloses the existential quantifier.

30 Extensionality

Definition 30.1. (Extension) The extension of a predicate is all the things that the predicate is true of.

Example 30.2. FOL is an extensional language because it does not represent differences of meaning between predicates that have the same extension (this is why we say we symbolize English into FOL rather than translate into FOL).

Example 30.3. (Extensionality)

- For any well-defined group of objects, we can define a one-place predicate that is true only for items in that group (i.e. that group is the extension)
- If a predicate is not true of anything, we say that its extension is empty, You can define a predicate to simply be empty.
- This logic can be extended to multi-place predicates by using set theory and considering ordered sets of names

Definition 30.4. (Interpretation) An interpretation in FOL (contrast to valuation in TFL) consists of the following:

1. The specification of a non-empty domain
2. Each sentence letter we care about is assigned a truth value
3. Each name that we care to consider is assigned exactly one object within the domain (its referent)
4. For each predicate that we consider (apart from '='), a specification of what things (in what order) the predicate is true of (its extension)

31 Truth in FOL

Definition 31.1. (Truth of Atomic Sentences) An atomic sentence $\mathcal{R}(a_1, a_2, ..., a_n)$ is true in an interpretation iff $\mathcal{R}$ is true of the objects named $a_1, a_2, ..., a_n$ (in that order) in the interpretation.

Definition 31.2. (Truth of Identity Predicate) The sentence $a = b$ is true if and only if a and b refer to the same object in the given interpretation.

Definition 31.3. (Truth of Sentential Connectives) The truth of sentential connectives follow the usual rules (as defined in TFL and FOL), the truth or falsehood must be determined given the interpretation though.

Definition 31.4. (Truth of Quantifiers) The truth of quantifiers is determined by the following process:

1. Starting with a formula with some free variable, say x in $A(...x...x...)$, replace all instances of x with c where c is a name to get $A(...x...x...)$
   (a) The resulting formula is called substitution instance of $\forall x\mathcal{A}$ or $\exists x\mathcal{A}$
   (b) The name c is called the instantiating name (x is the instantiated variable)
2. If $\mathbf{I}$ is an interpretation then $\mathbf{I}[d/c]$ is the same interpretation as I, but with some object in the domain d assigned to the name c (assumed to not already be used)
3. Then an object d satisfies $A(...x...x...)$ in interpretation $\mathbf{I}$ iff $A(...c...c...)$ is true in $\mathbf{I}[d/c]$ (assuming c does not already occur in $A(...c...c...)$)

Extending this definition to the quantifiers $\forall$ and $\exists$ we find that:
Definition 31.5. (Truth of Quantifiers)

- $\forall xA(...x...x...)$ is true in an interpretation iff every object in that domain satisfies $A(...x...x...)$
- $\exists A(...x...x...)$ is trye in an interpretation iff at least one object in the domain satisfies $A(...x...x...)$

32 Semantic Concepts

Definition 32.1. (Entailment) We say the sentences $\mathcal{A}_\infty, \mathcal{A}_\in, ..., \mathcal{A}_\setminus$ entail sentence C iff there is no interpretation for which $\mathcal{A}_\infty, \mathcal{A}_\in, ..., \mathcal{A}_\setminus$ are true and C is false.

- In this case we write $\mathcal{A}_\infty, \mathcal{A}_\in, ..., \mathcal{A}_\setminus \vDash \mathcal{C}$

Definition 32.2. (Validity) A sentence $\mathcal{A}$ in FOL is a validity iff A is true in every interpretation, i.e. $\vDash \mathcal{A}$

Definition 32.3. (Contradition) A sentence $\mathcal{A}$ in FOL is a contradiction iff A is false in every interpretation, i.e. $\vDash \neg\mathcal{A}$

Definition 32.4. (Valid in FOL) $\mathcal{A}_\infty, \mathcal{A}_\in, ..., \mathcal{A}_\setminus \therefore \mathcal{C}$ is valid in FOL iff there is no interpretation for which all of the premises are true and the conclusion is false, i.e., $\mathcal{A}_\infty, \mathcal{A}_\in, ..., \mathcal{A}_\setminus \vDash \mathcal{C}$

- If this is not the case, we say that $\mathcal{A}_\infty, \mathcal{A}_\in, ..., \mathcal{A}_\setminus \therefore \mathcal{C}$ is <u>invalid</u>

Definition 32.5. (Equivalent) Two sentences in FOL are equivalent iff they are true in exactly the same interpretations as each other; i.e., both $\mathcal{A} \vDash \mathcal{B}$ and $\mathcal{B} \vDash \mathcal{A}$.

Definition 32.6. (Jointly Satisfiable) The FOL sentences $\mathcal{A}_\infty, \mathcal{A}_\in, ..., \mathcal{A}_\setminus$ are said to be jointly satisfiable iff some interpretation makes them all true.

- If no such interpretation exists then these sentences are said to be <u>jointly unsatisfiable</u>.

33 Using Interpretations

Example 33.1. (Disproving Validity vs. Contradiction)

- To show that $\mathcal{A}$ is not a validity it suffices to find an interpretation for which $\mathcal{A}$ is false
- To show that $\mathcal{A}$ is not a contradiction it suffices to find an interpretation for $\mathcal{A}$ is true

Example 33.2. (Disproving Equivalence) To show that A and B are not equivalent it suffices to find an interpretation for which one is true and the other is false.

Example 33.3. (Satisfiability) Consider the FOL argument: $\exists x(G(x) \to G(a)) \therefore \exists x G(x) \to G(a)$ and the following interpretation:

- domain: Karl Marx, Ludwig von Mises
- G(x) : $_____x$ hated communism
- a: Karl Marx

The premise must be true since $G(a)$ is false so $G(a) \to G(a)$ is true so that $\exists(G(x) \to G(a))$ is true. On the other hand the conclusion is false since $\exists x G(x)$ is true (von Mises hated communism) but $G(a)$ is false (Marx did not hate communism). So the conclusion is false and the argument is invalid.

Definition 33.4. (Validity Proofs) If some interpretation makes $\mathcal{A}_\infty, \mathcal{A}_\in, .., \mathcal{A}_\setminus$ true and $\mathcal{C}$ false then:

1. $\mathcal{A}_\infty, \mathcal{A}_\in, .., \mathcal{A}_\setminus \therefore \mathcal{C}$ is invalid
2. $\mathcal{A}_\infty, \mathcal{A}_\in, .., \mathcal{A}_\setminus \nvDash \mathcal{C}$
3. $\mathcal{A}_\infty, \mathcal{A}_\in, .., \mathcal{A}_\setminus, \neg\mathcal{C}$ are jointly satisfiable

34 Reasoning about Interpretations

Example 34.1. (FOL Proof Concepts Table)

Term	**Yes (Prove)**	**No (Disprove)**
Validity?	all interpretations	One counter-interpretation
Contradiction?	all interpretations	One counter-interpretation
Equivalent?	all interpretations	One counter-interpretation
Satisfiable?	One counter-interpretation	all interpretations
Valid?	all interpretations	One counter-interpretation
Entailment?	all interpretations	One counter-interpretation

35 Properties of Relations

Example 35.1. (Transitivity) Consider the predicate $R(x, y)$: $_____x$ is older than $_____y$. The argument:

- Rudolf is older than Kurt. Kurt is older than Julia. Therefore, Rudolf is older than Julia is valid in english
- But $R(x, y)$, $R(y, z) \therefore R(x, z)$ is not valid in FOL because it relies on the transitivity of the predicate R (if A is older than B and B is older than C, then A is older than C).

When an argument depends on a property of a relation involved and that relation can be symbolized in FOL, then the symbolization of that property can be added to the argument to make it valid.

Definition 35.2. (FOL Argument Properties):

- Transitivity: A relation R is transitive iff it is the case that whenever $R(x, y)$ and $R(y, z)$ then $R(x, z)$
- Reflexive: A relation R is reflexive iff for any x, $R(x, x)$ holds
- Symmetric: A relation R is symmetric iff whenever $R(x, y)$ holds, so does $R(y, x)$
- Anti-Symmetric: A relation R is anti-symmetric iff for no two different x and y $R(x, y)$ and $R(y, x)$ both hold

Example 35.3. (Symbolizing Properties)

- Transitivity: If R is transitive then $\forall x \forall y \forall z((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$
- Reflexive: If R is reflexive then $\forall x R(x, x)$ holds
- Symmetric: If R is symmetric then $\forall x \forall y(R(x, y) \rightarrow R(y, x))$
- Anti-Symmetric: If R is anti-symmetric then $\neg \exists x \exists y((R(x, y) \wedge R(y, x)) \wedge \neg x = y)$ or equivalently $\forall x \forall y(R(x, y) \wedge R(y, x)) \rightarrow x = y$

Definition 35.4. (Equivalence Relation) A relation that is symmetric, reflexive, and transitive is called an equivalence relation

Definition 35.5. (Partial Orders) A relation that is reflexive, transitive and anti-symmetric, is called an partial order (e.g. less than or equal to and greater than or equal to)

36 Basic Rules for FOL

Definition 36.1. (Universal Elimination, pg. 298) If you have line $\forall x\mathcal{A}(...x...x...)$ on any line then you can get $\mathcal{A}(...c...c...)$ on any subsequent lines where c is any name.

- Note: the above means you replace ALL free instances of x in $\mathcal{A}(...x...x...)$ (ignoring the $\forall$ which must be the main logical operator) with the name c. (i.e., a substitution instance)

MAKE SURE TO WRITE [$\forall E$, m] WHERE m IS THE LINE OF THE UNIVERSAL QUANTIFIER

Definition 36.2. (Existential Introduction, pg. 299) If you have line $\mathcal{A}(...c...c...)$ where c is a name then you can get $\exists\mathcal{A}(...x...c...)$ on any subsequent lines where x is any variable.

- Note: the above means you replace some or all instances of name c in $\mathcal{A}(...c...c...)$ with the variable x.

MAKE SURE TO WRITE [$\exists I$, m] WHERE m IS THE LINE OF THE EXISTENTIAL QUANTIFIER

Note: The above rules always imply that something must exist in the domain in question (or else there would be a vacuous quantification). Hence the rule that every domain must have at least on object.

Definition 36.3. (Universal Introduction, pg. 302) If you have line $\mathcal{A}(...c...c...)$ where c is a name then you can get $\forall x\mathcal{A}(...x...x...)$ on any subsequent lines where x is any variable. The name c must not exist in any premises or undischarged assumptions.

- Note: the above means you replace ALL instances of name c in $\mathcal{A}(...c...c...)$ with the variable x.

MAKE SURE TO WRITE [$\forall I$, m] WHERE m IS THE LINE OF THE UNIVERSAL QUANTIFIER

Definition 36.4. (Existential Elimination, pg. 306) If you have the line $\exists x\mathcal{A}(...x...x...)$ and you can use $\mathcal{A}(...c...c...)$ to derive $\mathcal{B}$ in a subproof then you can obtain $\mathcal{B}$ on any subsequent lines. Note: c must not occur in any undischarged assumptions before the subproof, or inside $\exists x\mathcal{A}(...x...x...)$, or inside the conclusion $\mathcal{B}$ of the subproof.

MAKE SURE TO WRITE [$\exists E$, m, i-j] WHERE m IS THE LINE OF THE EXISTENTIAL QUANTIFIER and i-j ARE THE LINES OF THE SUBPROOF. MAKE SURE TO WRITE [AS] FOR EACH SUBPROOF ASSUMPTION.

37 Proofs with Quantifiers

Example 37.1. (Constructing Proofs with Quantifiers) Two strategies for working with quantifiers are to work either forwards or backwards from them. Only work backwards from existentially quantified sentences or work forwards from universally quantifed sentences when all other strategies have been applied. See Chapter 37 for more details.

38 Conversion of Quantifiers

Definition 38.1. (Conversion of Quantifiers First Rule) Note that $\forall x \neg \mathcal{A}$ and $\neg \exists x \mathcal{A}$ are equivalent and can be obtained from each other.

MAKE SURE TO WRITE [CQ, m] WHERE m IS THE LINE OF THE SENTENCE BEING CONVERTED

Definition 38.2. (Conversion of Quantifiers Second Rule) Note that $\exists x \neg \mathcal{A}$ and $\neg \forall x \mathcal{A}$ are equivalent and can be obtained from each other.

MAKE SURE TO WRITE [CQ, m] WHERE m IS THE LINE OF THE SENTENCE BEING CONVERTED

39 Rules for Identity

This section is N/A.

40 Derived Rules

This section is N/A.

41 Proofs and Semantics

Example 41.1. (Relationship between $\vdash$ and $\models$) It is important to note that $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n \vdash \mathcal{C}$ (proof-theoretic) and $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n \models \mathcal{C}$ (semantic) are two different concepts. The first states that there is a proof with no undischarged assumptions that has $\mathcal{A}_n$ as premises and ends with $\mathcal{C}$ as the conclusion. The second states that there is not valuation that makes $\mathcal{A}_n$ true and $\mathcal{C}$ false.

Definition 41.2. (Equivalence of Provability and Entailment) $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n \vdash \mathcal{C}$ iff $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n \models \mathcal{C}$

Example 41.3. (Consequences of Extensional Equivalence of Provability and Entailment)

- An argument is valid if and only if the conclusion can be proved from the premises
- A sentence is a validity if and only if it is a theorem
- Two sentences are equivalent if and only if they are interderivable
- Sentences are jointly satisfiable if and only if they are jointly consistent

Example 41.4. (FOL Proof Concepts Table)

Question	**Yes (Prove)**	**No (Disprove)**
Is $\mathcal{A}$ a validity / theorem?	Prove that $\vdash \mathcal{A}$	Find an interpretation for which $\mathcal{A}$ is false
Is $\mathcal{A}$ a contradiction?	Prove $\mathcal{A} \vdash \bot$	Give an interpretation for which $\mathcal{A}$ is true
Are $\mathcal{A}$ and $\mathcal{B}$ equivalent?	Prove $\mathcal{A} \vdash \mathcal{B}$ and $\mathcal{B} \vdash \mathcal{A}$	Give an interpretation for which $\mathcal{A}$ and $\mathcal{B}$ have different truth values
Are $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n$ jointly satisfiable?	Find and interpretation for which $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n$ are all true	Prove $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n \vdash \bot$
Is $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n \therefore \mathcal{C}$ valid?	Prove $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n \vdash \mathcal{C}$	Find an interpretation for which Are $\mathcal{A}_1, \mathcal{A}_2, ..., \mathcal{A}_n$ are true and $\mathcal{C}$ is false